

Project Final Report

1. Changes in Project Directions

Original Proposal:

- The project aimed to document criminal activity at different universities across the Midwest using data from university police departments. It intended to showcase where and when crimes occurred across these universities to help students and faculty stay informed and safe. It initially intended to focus on universities in the Midwest like University of Illinois at Urbana-Champaign, University of Michigan, and Michigan State University.

Final Project:

- The focus shifted from tracking crime trends to providing better insights using crime statistics. Location and time were no longer separate entities but were incorporated under the crime table. The geographical coverage was extended to include universities outside the Midwest. The full list of universities used ended up being University of Illinois at Urbana-Champaign, University of Nebraska-Lincoln, University of Florida, and Michigan State University.

2. Application Achievements and Failures

Achievements:

- Successfully listed crimes effectively.
- Compiled different crime logs from different universities into one larger database
- Linked crime data with city demographics to gain more insight into correlations between crimes and demographics
- Completed all CRUD functionalities with filters and searches.
- Provided users with better insights compared to the daily crime log, aiding awareness of crime patterns.

Failures:

- The map function could not be effectively implemented due to roadblocks with data parsing and calculating distance differences.

3. Changes in Data Schema or Sources

Data Source Changes:

- Used crime statistics tables from across the country and states for more accurate insights into general crimes.

- Focused on data from four universities: UIUC, University of Nebraska-Lincoln, University of Florida, and Michigan State University. Removed University of Michigan because its format for crime logs was inconsistent with the other universities.

Schema Changes:

- Incorporated feedback to move time and location under the crime table instead of keeping them as separate entities.
- Added CrimeStatistics and City entities for better data representation.
- Assigned the crime log dataset as a weak entity that can not exist without its corresponding university.

4. UML Diagram and Table Implementation Changes

Original Design:

- Entities included Crime, University, User, Location, and Time.

Final Design:

- Time and location were moved under the crime table.
- Added CrimeStatistics and City entities. The CrimeStatistics entity provided general information about the crime type (for example: average age of perpetrators of theft across the United States). The city entity provided demographics data for the location (for example: population, median age, poverty rate).
- Crime changed to a weak entity that can only be uniquely identified by its crimeId and university it took place in.

Reasons for Changes:

- Location and time provided more insights when included under the crime table.

Changes in ER Diagram and Table Implementation:

- **Original ER Diagram:**
 - Entities: Crime, University, User, Location, Time.
 - Relationships: University - Crime (Logs), University - City (Located), City - Crime (Occurred In), Crime - CrimeStatistics (Related), Crime - Users (Proximity).
- **Final ER Diagram:**
 - Entities: Crime, University, User, City, CrimeStatistics.
 - Relationships: University - Crime (Logs), University - City (Located), City - Crime (Occurred In), Crime - CrimeStatistics (Related), Crime - Users (Proximity).

Detailed Entity Descriptions:

- **Crime:**
 - Attributes: crimeId (PK), crimeType, crimeAddress, occurredDate, occurredTime, reportedTime, status, universityName.
- **University:**
 - Attributes: universityId(PK), universityName.
- **Users:**
 - Attributes: username (PK), password, userLocation.
- **City:**
 - Attributes: cityName (PK), population, medianAge, povertyRate, medianIncome, medianProperty.
- **CrimeStatistics:**
 - Attributes: crimeType (PK), perpetratorAvgAge, victimAvgAge.

5. Functionalities Added or Removed

Added:

- Enhanced CRUD functionalities with filters and searches.
- User Login page.
- A crime logs page with filters for user-friendly viewing of data.

Removed:

- The map function due to technical difficulties in parsing location data and calculating distance differences.

6. Advanced Database Programs

Complementary Features:

- Advanced SQL queries and indexing significantly improved performance for filtering and joining data.
- Stored procedures handled complex data manipulations and aggregations effectively.

7. Technical Challenges

Challenge 1: Parsing and Normalizing Location Data

Description:

- **Problem:** We faced significant challenges in parsing and normalizing location data to calculate distance differences between crime locations. The raw location data provided by different universities was inconsistent and often lacked uniform

formatting. This inconsistency made it difficult to standardize the data for accurate distance calculations.

- **Impact:** Without accurate geolocation data, the map feature that was initially planned to visually display crime locations and calculate distances between incidents could not be implemented effectively.

Solution:

- **Approach:** We explored using advanced geolocation APIs and algorithms that could convert raw address data into precise latitude and longitude coordinates. These APIs could then be used to calculate distances between crime locations accurately.
- **Implementation:** Given more time, we would integrate services like Google Maps Geocoding API or similar tools early in the development process to ensure smooth and accurate geolocation data processing.

Advice:

- **Integration Timing:** Integrate geolocation services early in the development process to ensure consistent data formatting and accurate distance calculations.
- **Data Standardization:** Establish a standardized format for location data from the outset to avoid complications later in the project. Use robust data validation techniques to ensure all location data adheres to this format.
- **API Selection:** Select geolocation APIs that offer both accuracy and efficiency, and ensure they can handle the expected volume of data requests.

Challenge 2: Handling Large Datasets Efficiently

Description:

- **Problem:** Managing and querying large datasets efficiently while maintaining performance was a significant challenge. As the project expanded to include more universities and a larger volume of crime data, the performance of our database queries began to degrade.
- **Impact:** Slow query performance affected the user experience, particularly when performing complex searches or generating reports on crime data.

Solution:

- **Approach:** To improve performance, we implemented indexing on key columns and optimized our SQL queries. Indexing helped speed up data retrieval by allowing the database to quickly locate the rows that match the query conditions.
- **Implementation:** We used the EXPLAIN ANALYZE command to measure the performance of our queries before and after adding indices. This helped us identify the most effective indexing strategies and optimize our queries accordingly.

Advice:

- **Performance Analysis:** Regularly use performance analysis tools like EXPLAIN ANALYZE to identify bottlenecks in your queries.
- **Indexing:** Implement indexing on frequently queried columns to improve query performance. Consider the trade-offs between read and write performance when deciding where to apply indices.
- **Query Optimization:** Continuously optimize SQL queries to ensure they run efficiently. Avoid complex joins and subqueries where possible, and use appropriate SQL functions to streamline data retrieval.

Challenge 3: Ensuring Data Consistency and Integrity

Description:

- **Problem:** Ensuring data consistency and integrity across multiple data sources was a complex challenge. The crime data came from various universities, each with different reporting standards and formats. For example, when trying to connect city demographic data with data from university crime logs, they wrote the state name in different formats (Illinois vs IL). This variability introduced risks of data inconsistencies and inaccuracies.
- **Impact:** Inconsistent data could lead to incorrect analyses and unreliable insights, undermining the application's usefulness and credibility.

Solution:

- **Approach:** We used transaction management and data validation techniques to maintain data consistency and integrity. Transactions ensured that database operations were completed successfully or rolled back entirely in case of errors. Data validation techniques were employed to verify the accuracy and consistency of incoming data. Inconsistent formatting for data could also be fixed by opening the CSV file in a text editing software like excel and manually changing the data format to match across all sources.
- **Implementation:** Transactions were implemented with the appropriate isolation levels to prevent data anomalies. Validation scripts checked for common issues such as missing or incorrectly formatted data before it was inserted into the database. For the inconsistent state names, we opened the CSV files in excel and used find and replace to change the state abbreviations to full state names.

Advice:

- **Transaction Management:** Use transactions to ensure that database operations are atomic, consistent, isolated, and durable (ACID properties). This helps maintain data integrity even in case of errors or interruptions.

- **Data Validation:** Implement rigorous data validation processes to detect and resolve inconsistencies early. Validate data at multiple stages: before insertion, during processing, and before generating reports.
- **Standardization:** Standardize data formats and reporting standards across all data sources. Establish clear guidelines for data collection and reporting to minimize discrepancies.

8. Other Changes

Data Source Integration:

- **Original Proposal:** The application was intended to use CSV data from a limited number of university police departments, primarily focusing on the last 60 days of crime logs.
- **Final Application:** Expanded to incorporate more comprehensive crime statistics from various universities and states, allowing for a broader and more detailed dataset. This change provided users with a richer set of data for analysis.

9. Future Work

Improvements:

- Implement the map feature more effectively with better data parsing and geolocation algorithms.
- Enhance the user interface for more intuitive and user-friendly interactions.
- Integrate real-time data updates for more current information.

10. Division of Labor and Teamwork

Final Division of Labor:

- **Mithesh:** Handled tables connecting different colleges, wrote sql queries, indexing
- **Albert:** Handled tables connecting different crimes, wrote sql queries, indexing
- **Teja:** Handled tables connecting crime and crime statistics and the logs page, wrote queries, importing data, frontend

Teamwork:

- The team managed to collaborate effectively, with clear communication and task distribution ensuring the project was completed on time. Regular meetings and feedback sessions helped address any issues promptly.

Throughout the development of our project, we barely relied on ChatGPT for assistance with our front-end code. While ChatGPT provided occasional guidance and suggestions, the majority of the code was independently developed by our team. We utilized our own knowledge and skills to integrate the user interface with the backend database, ensuring seamless data retrieval and storage. The minimal use of ChatGPT was primarily focused on resolving specific technical issues and optimizing certain aspects of our code. This approach allowed us to gain a deeper understanding of the project and maintain full control over the development process. We used Project Track 1 Workshop 3 as a source to base our website's backend and some frontend pages off of.